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Original Research

Correcting for the Echo-Time Effect After Measuring the Cerebral Blood Flow by Arterial Spin Labeling

Jack R. Foucher, MD, PhD,[1–3]* Daniel Roquet, MS,[1,2] Corinne Marrer, MT,[1,4] Bich-Thuy Pham, RA,[1,2] and Daniel Gounot, PhD[1,3,4]

Purpose: To take into account the echo time (TE) influence on arterial spin labeling (ASL) signal when converting it in regional cerebral blood flow (rCBF). Gray matter ASL signal decrease with increasing TE as a consequence of the difference in the apparent transverse relaxation rates between labeled water in capillaries and nonlabeled water in the tissue (δR_2^*). We aimed to measure ASL/rCBF changes in different parts of the brain and correct them.

Materials and Methods: Fifteen participants underwent ASL measurements at TEs of 9.7–30 ms. Decreases in ASL values were localized by statistical parametric mapping. The corrections assessed were a subject-per-subject adjustment, an average δR_2^* value adjustment, and a two-compartment model adjustment.

Results: rCBF decreases associated with increasing TEs were found for gray matter and were corrected using an average δR_2^* value of 20 s^{-1}. Conversely, for white matter, rCBF values increased with increasing TEs ($\delta R_2^* = -23\ s^{-1}$).

Conclusion: Our correction was as good as using a two-compartment model. However, it must be done separately for the gray and white matter rCBF values because the capillary R_2^* values are, respectively, larger and smaller than those of surrounding tissues.

Key Words: apparent transverse relaxation time; arterial spin labeling; echo time; regional cerebral blood flow; gray matter; white matter
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ARTERIAL SPIN LABELING (ASL) has gained popularity as a noninvasive method for regional cerebral blood flow (rCBF) quantification (1). It requires neither a radioactive tracer nor a contrast agent, and is now used routinely with clinical scanners.

All ASL methods subtract control images (no tagging) from those acquired after tagging the blood spins and are generally recorded at neck level. Labeled arterial spins flowing through the imaged slice cause a small signal decrease (~1%) revealed by the subtraction. ASL sequences are grouped as continuous, pulsed, and velocity sensitive (2,3). The pulsed sequence is the only one that is routinely used. Compared with continuous methods, pulsed methods do not require a special tagging coil and deposit smaller amounts of energy in the tissue, which is paid for by an ~35% signal reduction (4). An increase in the scanner field strength (from 1.5 to 3 Tesla [T]) compensates in part for the signal reduction because the blood spins longitudinal relaxation time T_1 increases (1200 to 1500 ms) although the higher field strength should be used with methods that deposit small amounts of energy (1,3).

ASL measurements can be combined with blood-oxygenation-level-dependent (BOLD) measurements to reduce the scan time and allow the (perfect) superposition of the two measurements. However, there is a reduction in the ASL signal, which is proportional to the echo time (TE) (5), and it is then difficult to compare rCBF-converted images acquired at different TEs (6). Manufacturers have suggested an rCBF conversion equation (7), but it does not correct for TE differences, and thus results cannot be compared for different TEs.

The purpose of this study was to quantify the TE effect on rCBF evaluations and compare two equations that might correct the evaluations, one of which is simple, whereas the other is more complicated. The simpler equation merely weighs the equation proposed by Wang and colleagues (7) for the TE effect using an exponential correction term. The more sophisticated algorithm is the two-compartment model of St Lawrence and Wang (6) with model parameters adapted for a field strength of 3T. T_2^* values of the gray matter and venous blood compartments were directly measured. Fifteen subjects took part in the study.

[1]UdS, Université de Strasbourg, Strasbourg, France.
[2]INSERM U666—Physiopathologie et psychopathologie cognitive de la schizophrénie, Strasbourg, France.
[3]HUS, Hôpitaux Universitaires de Strasbourg, Strasbourg, France.
[4]LINC, Laboratoire d'Imagerie et de Neurosciences Cognitives, CNRS-UdS, Strasbourg, France.
Additional supporting information may be found in the online version of this article.
*Address reprint requests to: J.R.F., UdS/HUS, Service de Physiologie, Clinique Psychiatrique, INSERM U666, Hôpitaux Universitaires, BP 406, 67091 Strasbourg Cedex, France.
E-mail: jack.foucher@laposte.net
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THEORY

The ASL to CBF conversion was performed according to the following equation derived from the classic Kety-Schmidt single-compartment model (7–9, see Wang et al for a complete theoretical development and the associated appendices for theoretical details) (4):

$$rCBF = \frac{\lambda\,\Delta M}{2\alpha M_0 \cdot TI_1 \cdot e^{-TI_2/T_{1a}}} \qquad [1]$$

where ΔM is the magnetization difference between the tagged and nontagged images that arises because of the blood signal reversal. λ is the brain/blood partition coefficient for water (we used an average value of 0.9 mL/g, although the value is 0.98 for gray matter and 0.81 for white matter) (10). λ can be interpreted as the volume in which the tracer distributes, i.e., labeled water per unit weight of tissue. α is the inversion efficiency (0.95). M_0 is the tissue magnetization (measured using the first echo planar image that was not tagged). TI_1 is the duration between the inversion and the saturation pulse (600 ms). TI_2 is the duration between the inversion and image acquisition, which ranged from 1325/2990 to 2421/2990 ms, depending both on the image number and the TE. T_{1a} is the longitudinal relaxation time for blood (1500 ms at 3T). According to the authors, this equation is only valid within a limited time window, i.e., TI_2 is larger than the transit time from the tagged region to the capillaries in the slice but is smaller than the time from the tagged region to the gray matter, i.e., the former time plus the exchange time (4). Indeed, the latter is not negligible (11).

There are three components in the equation: the efficiency and tagging duration (α and TI_1, respectively), the resulting signal ($\lambda\Delta M/M_0$), and the signal's modulation by T_1, i.e., the signal decrease caused by the time it takes for the tag to reach the brain (where it is recorded) plus the image acquisition time ($\exp(-TI_2/T_{1a})$). The acquisition time signal, ΔM, is normalized relative to M_0, i.e., expressed as the fraction of labeled water, which acts as the tracer, relative to the local water and is converted into mL/g gray matter using λ. However, whereas M_0 is modulated by the T_2^* of the whole tissue, i.e., in our case, the gray matter (T_{2GM}^* or its inverse, R_{2GM}^*, i.e., $\exp(-R_{2GM}^* \times TE)$) and the white matter ($T_{2WM}^*$ or its inverse, R_{2WM}^*), ΔM is modulated by the T_2^* of capillary blood (Cap), which has a value between those of arterial and venous blood (T_{2Cap}^* or its inverse, R_{2Cap}^*, i.e., $\exp(-R_{2Cap}^* \times TE)$). R_{2Cap}^* is expected to be larger than R_{2GM}^*.

Accordingly, to correct for the effect of the T_2^* difference between capillary blood and gray matter at different TE, and to determine the magnetization in the absence of it, the second component of Eq. [1] is written as:

$$\frac{\lambda\,\Delta M/e^{-R_{2Cap}^*\cdot TE}}{M_0/e^{-R_{2GM}^*\cdot TE}} = \frac{\lambda\,\Delta M \cdot e^{(R_{2Cap}^*-R_{2GM}^*)\cdot TE}}{M_0} = \frac{\lambda\,\Delta M \cdot e^{\delta R_2^*\cdot TE}}{M_0}$$

with $\delta R_2^* = R_{2Cap}^* - R_{2GM}^*$. R_{2GM}^* can be measured, but R_{2Cap}^* is not measurable and depends on the partial desaturation of blood oxygen, which adds a BOLD effect to the ASL measurements. However, δR_2^* can be directly assessed by measurement of the decrease in the ASL or rCBF value caused by the TE effect and can then be compared with expected values (see below).

Thus, a simple modification of Eq. [1] takes into consideration the TE effect (5):

$$rCBF = \frac{\lambda\,\Delta M e^{\delta R_2^*\cdot \mathrm{TE}}}{2\alpha M_0 \cdot TI_1 \cdot e^{-TI_2/T_{1a}}} \qquad [2]$$

The TE effect is best taken into account using the two-compartment model of St Lawrence and Wang (6). This model includes contributions from capillary blood (ΔM_{Cap}) and gray matter (ΔM_{GM}) to the ASL signal (ΔM):

$$\begin{aligned}\Delta M &= \Delta M_{Cap} + \Delta M_{GM}\\ \Delta M &= cst_{Cap} \times CBF + cst_{GM} \times CBF\\ &= CBF(cst_{Cap} + cst_{GM})\\ CBF &= \Delta M/(cst_{Cap} + cst_{GM}),\end{aligned}$$

where CBF is the whole cerebral blood flow, cst_{Cap} is a value that accounts for the contribution of the capillary blood to the ASL signal, and cst_{GM} is a similar correction used for the gray matter ASL signal. This model also takes into consideration that as TI_2 increases, the T_2^* value for the capillary blood flow approaches that for venous blood. Because our TI_2 values were larger than the capillary transit time, we used Eqs. [9] and [13] given by St Lawrence and Wang (6) (which are referred to in this article as Eq. [3]). The equations and the values of the constants adjusted for 3T are given here (we refer the reader to reference 6 for a complete theoretical development):

$$cst_{Cap} = \left(\frac{2M_0}{\lambda}\right)\frac{e^{-TE\cdot R_{2a}^*}e^{-R_{1a}\cdot TI_2}}{(PS/V_c + TE\cdot\eta)}\left\{1 - e^{-(PS/F + TE\cdot(R_{2v}^* - R_{2a}^*))}\right\}$$

with $\eta = (R_{2v}^* - R_{2a}^*)/\tau_c$ and $\tau_c = V_c/CBF$

$$cst_{GM} = \left(\frac{2M_0}{\lambda}\right)\frac{e^{-TE\cdot R_{2b}^*}}{\delta R_1}\left\{\beta E_R e^{-R_{1b}TI_2} E e^{-R_{1a}TI_2}\right\}$$

with $\delta R_1 = R_{1a} - R_{1b}$, $E = 1 - e^{-PS/CBF}$, $E_R = 1 - e^{-(PS/CBF\ -\ \delta R_{1\cdot\tau c})}$, and $\beta = \left(1 + \frac{\delta R_1 V_c}{PS}\right)^{-1}$ where PS is the capillary permeability-surface area product (140 mL/100 g/min), Vc the distribution volume of tracer in capillary space (2 mL/100 g), the Rs are the inverse of arterial T_1 (T_{1a} = 1500 ms), brain T_1 (T_{1b} = 1250 ms), and arterial (T_{2a}^* = 97 ms), brain (T_{2b}^* = 61 ms) and venous T_2^* (T_{2v}^* = 10 ms).

MATERIALS AND METHODS

Subject and Task

Fifteen subjects were included (mean age 39 ± 10 years old, age range 26–63 years old, 8 women). None had a contra-indication(s) for MRI and were free of any neurological or psychiatric illness. The study was part of an optimization protocol approved by the local ethical committee, and the subjects signed an informed consent form. For all sessions, the subjects remained still without falling asleep.

Data Acquisition

Images were collected using a Siemens Verio 3T scanner equipped with a 32-channel head coil (Erlangen, Germany). For all pulsed ASL experiments, the QUIPPS II sequence provided by the manufacturer was used (12). Parameters were: TR = 3 s, TE = 9.7–30 ms (see below), voxel size = 4 × 4 × 4 mm, to cover the whole brain without leaving a gap, 28 serial slices were obtained, 100 volumes, and an acceleration factor of 2 (Generalized Autocalibrating Partially Parallel Acquisition or GRAPPA). Following the terminology of Wong and colleagues (12), the ASL parameters were: TI_1 = 600 ms, TI_2 was between 1325/2421 to 1325/2990 ms, which depended on the number of images and the TE. The tagged volume was 10 cm thick, positioned at the neck, and its distal part was 23 mm below the first slice to avoid saturation. Bipolar gradients were used to eliminate the signals from fast moving spins (>10 cm/s), which probably arose from arterial blood flow (12,13). The first volume recorded corresponded to the M_0, and nontagged images were acquired in alternation with tagged images. Inhomogeneities were corrected using the filter built into the instrument.

The TE was varied from 9.7 to 30 ms in 3-ms steps. The session order was randomized. The number of scans (100 per 5-min session) was chosen because we observed that the reproducibility of the CBF/ASL results did not change much for the same setting and between settings when more than 100 scans were acquired.

Data Processing

The scanner DICOM Images were transformed into analyze format by MRI Convert (Lewis Center for Neuroimaging, University of Oregon, Eugene, OR) and processed using the SPM2 toolbox in Matlab 5.3 (MathWorks, Inc., Natick, MA) for image manipulation, registration, and segmentation and voxel-based statistical analysis (Functional Imaging Laboratory, Institute of Neurology, London, UK).

Because Eqs. [2] and [3] depend on T_2^*, we measured it for each subject. The signal decreases for the M0 images were modeled as a function of $\exp(-TE/T_2^*)$ with T_2^* values between 1 and 1000 ms to find the best fits between the experimental data and the model. The T_{2GM}^* was averaged for the GM images according to the segmentation performed on the proton density maps. We looked for a relationship between T_{2GM}^* and age, which could have altered the rCBF estimation. After normalization, the T_2^* maps for each subject were averaged, and the venous T_2^* was measured in the great cerebral vein of Galen.

After rigid registration of the echo planar images, the ASL signal was computed by averaging the differences between the tagged and corresponding nontagged images. The ASL to rCBF conversions were performed according to Eqs. [1], [2], and [3] using the parameters provided in the Supplementary Material, which is available online, that had been derived from the literature (6) or measured for this report.

To determine if the TE affects parts of the brain differently, we performed a statistical parametric mapping (SPM) analysis for the TE effect on the rCBF derived from Eq. [1]. Then, to determine if the images had been overcorrected or insufficiently corrected using the aforementioned compensatory coefficient, the SPM analysis was repeated using the corrected rCBF images.

For both SPM analyses, the rCBF images were first spatially normalized and then smoothed (6-mm 3D Gaussian kernel). We designed a matrix with a constant term plus one vector: $\exp(\delta R_2^* - TE)$ using δR_2^* derived previously. We computed a contrast map for each subject and then ran a second-level t-test on the contrast maps to perform a random effect analysis. Because we wanted to reduce type-2 errors, i.e., avoiding false negative voxels, the SPM results are presented with a threshold of t = 1.7, n = 15, $P \leq 0.05$, and were uncorrected for multiple testing.

RESULTS

For gray matter, the mean value for T_{2GM}^* was 61 ± 7 ms; for white matter, the mean value for T_{2WM}^* was 53 ± 3 ms; and for venous blood, the mean value for T_{2V}^* was 10 ±12 ms (at the MNI coordinate [0, −38, 2] for the great cerebral vein of Galen). There was no change in T_{2GM}^* with increasing age (r = −0.15, P = 0.59), whereas for gray matter, the mean rCBF depended on age (r = −0.53, n = 15, P = 0.02).

The mean rCBF values for gray matter are given for each TE with their associated errors (Fig. 1a). The mean rCBF values computed using Eq. [1] in its original form (gray) clearly decreased with increasing TE: from TE = 9.7 ms to 30 ms, there was a 35% decrease, i.e., $0.65 \exp(-\delta R_2^* \times 9.7 \times 10^{-3}) = \exp(-\delta R_2^* \times 30 \times 10^{-3})$, thus $\delta R_2^* = 21.2\ s^{-1}$. Using this principle for each TE, the mean δR_2^* value was 20 s^{-1} (range, 16–26 s^{-1}). The resulting mean rCBF value (for the data presented in black) was 62 ± 9.5 mL/100 g/min. When the rCBF values for the white matter were computed in the same manner using Eq. [1], there was a clear increase in the values (68% increase, Fig. 1c). The correction used a value of $\delta R_{2WM}^* = -23\ s^{-1}$ (range, 13–40 s^{-1}), which resulted in a calculated blood flow for the white matter of 12 ± 0.7 mL/100 g/min.

When the δR_2^* adjustment was done on a subject-by-subject basis, the mean δR_2^* was 22 ± 10 s^{-1}, and the mean rCBF for the gray matter values (dashed, triangle) was 64 ± 14 mL/100 g/min. There was no correlation between the ages of the subjects and their individual δR_2^* values (r = −0.005, not significant). When the two-compartment model was used (dashed, diamond), the mean value for the gray matter rCBFs was 69 ± 11 mL/100 g/min, and the results appeared well adjusted. Because the corrected data generally formed an inverted U-shaped curve, the

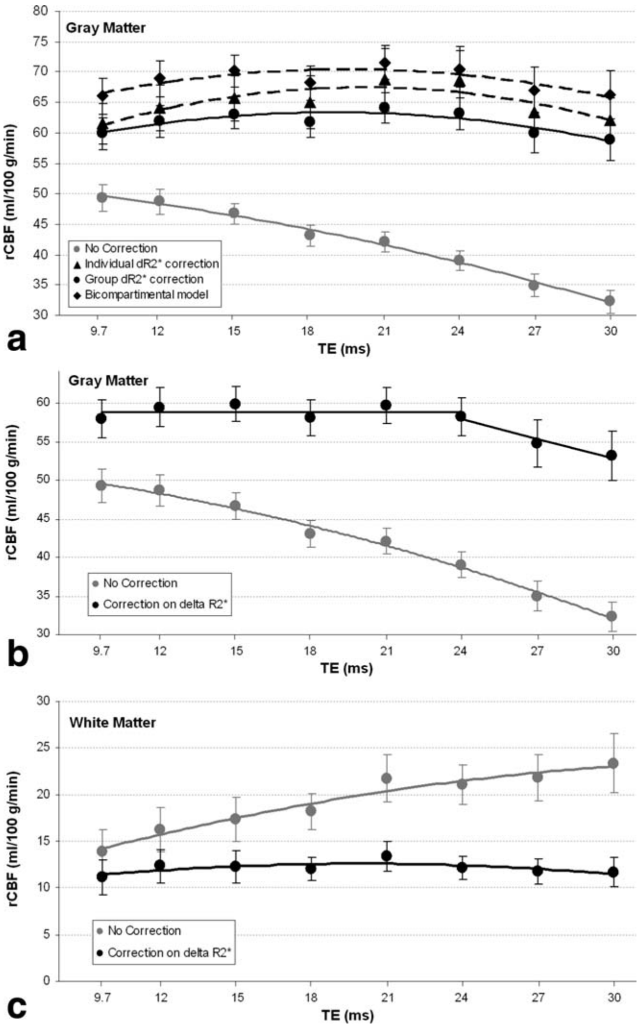

Figure 1. **a**: Regional cerebral blood flow (rCBF) averaged for all voxels classified as gray matter versus the acquisition echo time (TE) (mean ± SEM). The uncorrected values decrease exponentially (gray). Other values shown were corrected with a δR_2^* adjustment for each result (dashed, triangle), with a correction that used an averaged δR_2^* of 20 s^{-1} (black), or with the two-compartment model of St Lawrence and Wang (6) (dashed, diamond). All curve fitting used an exponent function. **b**: As a post hoc linear curve fitting using an averaged δR_2^* value of 16.7 s^{-1} (black), for TE values <27 ms, i.e., the plot was linear, and the data points had small SEMs. **c**: White matter rCBF values versus TEs, uncorrected (gray) and corrected values that used an averaged δR_2^* value of -23 s^{-1} (black).

curve fitting used a second-order polynomial equation (retrospective extrapolation would give rCBF = 45 mL/100 g/min at TE = 0 ms and prospective extrapolation an rCBF = 0 mL/100 g/min at TE = 55 ms). As a post hoc, fine-tuning curve fitting to obtain linearity, we used a value for δR_2^* of 16.7 s^{-1} between 9.7 and 24 ms for the gray matter rCBF values. For larger TEs, however, the slope was negative and the standard deviations were larger (Fig. 1b).

Given the values for δR_2^* of 20 s^{-1} and R_{2GM}^* of 17 s^{-1}, $R_{2Cap}^* = \delta R_2^* + R_{2GM}^* = 37$ s^{-1} (thus, the average capillary T_{2Cap}^* was 27 ms). When the same computation was performed for the white matter, given $\delta R_{2WM}^* = -23$ s^{-1} and $R_{2WM}^* = 19$ s^{-1}, $R_{2WM\_Cap}^* = -4$ s^{-1}.

When looking for regions affected by the TE, the gray matter was affected, whereas the white matter was not (Fig. 2, left). The TE did not have a large effect on cerebellum images, in contrast to its large effect on the images of the supratentorial structures. The TE effect was reversed for the white matter images in comparison with the gray matter images as the rCBF values increased with increasing TE, predominantly in the upper part of the brain. Finally, after correcting the CBF images, there was no noticeable TE effect anywhere in the cortex. Conversely, all white matter images were affected by a reverse TE effect (Fig. 2 right).

DISCUSSION

When uncorrected, rCBF values for gray matter decreased with increasing TE. Cerebellum rCBF values were less affected than were those of the supratentorial structures, and TE had the reverse effect on those of the white matter. It was sufficient to correct the rCBF values for the gray matter with $\exp(\delta R_2^* \times \text{TE})$ and use an averaged δR_2^* value. The correction eliminated the TE effect over the range of values, and the standard deviations for the rCBF values were the smallest for all of the corrections. The δR_2^* and T_{2GM}^* values were not age dependent, whereas the rCBF was. The average capillary T_{2Cap}^* value was 27 ms, which was closer to that for venous blood (10 ± 12 ms) than to the arterial value (~100 ms) (6).

Thus, it was possible to correct for the TE effect on the mean rCBF/ASL values with the simple modifier $\exp(\delta R_2^* \times \text{TE})$. For our normal population, the δR_2^* value for gray matter was ~20 s^{-1} and did not change with age. Additional studies are needed to be certain that this value is not different for the brains of Alzheimer's disease patients or those with disorders that arise as a consequence of iron deposits. Using our correction may be appropriate even when the TE is short, because the average population rCBF value then agrees more closely with positron emission tomography (PET) measurements, i.e., 55–75 mL/100 g/min (14–17).

Considering the corrected curves, it appears that when TE is >24 ms at 3 Tesla (T), gray matter rCBF values decrease as the TE increases, and the uncertainties in the values also increase. The reasons for these findings are unclear and remain to be elucidated, but one reason may be associated with the decrease in the signal-to-noise ratio. Indeed at longer TEs, the ASL signal decreases both because of the T1 effect caused by the increased imaging time (ΔTR = 569 ms; thus, a signal decreases 21% between TE values of 9.7 and 30 ms) and a larger T_2^* effect.

Notably, the average capillary T_{2Cap}^* is similar to the one for venous blood, suggesting that when a region is active the BOLD effect may dramatically decrease the δR_2^* value, which would lead to an overestimation of the ΔrCBF value between the active and the inactive states (5). Such an effect would increase as the

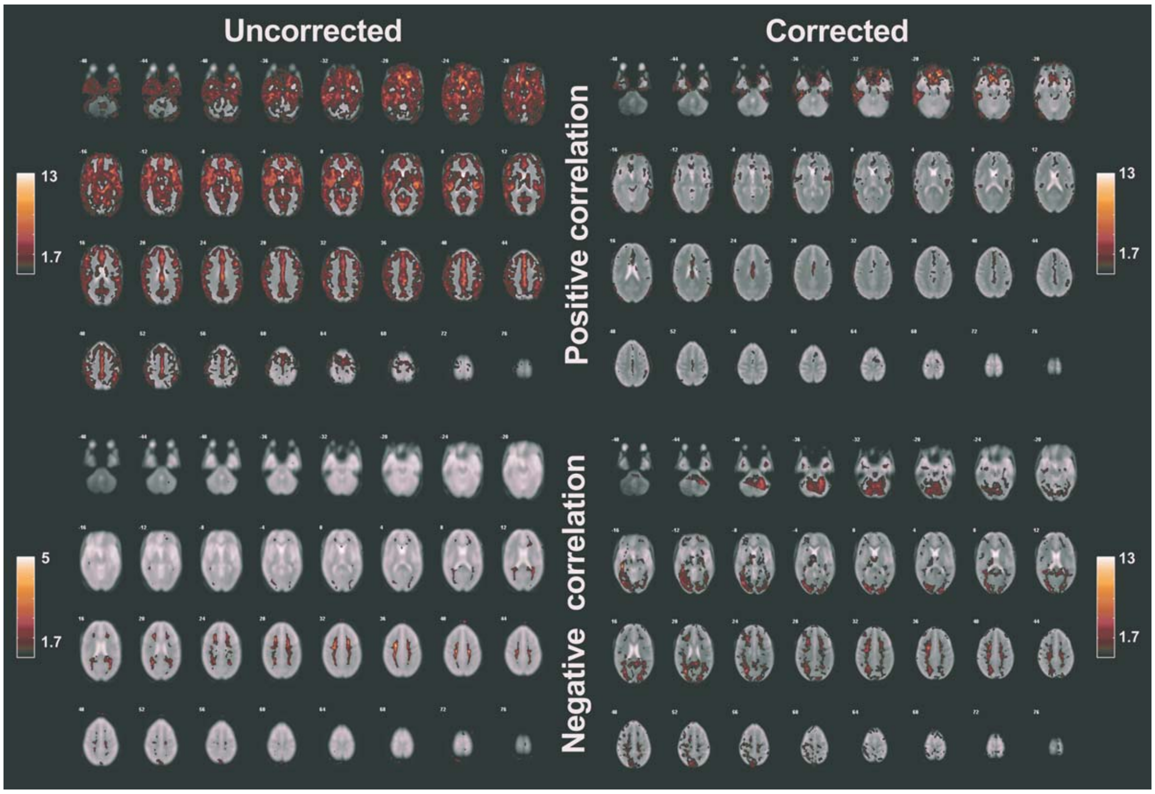

Figure 2. SPM results for the correlation between rCBF and exp($\delta R_2^* \times$ TE) ($P < 0.05$, extension > 50 voxels, i.e., 0.4 mL). Left panels: uncorrected images. With a few exceptions, the gray matter rCBF values correlated significantly with exp($\delta R_2^* \times$ TE) (hot colors in the upper left panels are the z-score coded according to the corresponding color bar), whereas many of white matter rCBF values were inversely correlated (hot colors in the lower left panels). For the corrected images, those for gray matter were no longer correlated with exp($\delta R_2^* \times$ TE) (hot colors in the upper right panels), whereas the negative effect for the white matter increased because the correction used was the gray matter value for δR_2^* (hot colors in the lower right panels). Numbers are the slice position in the z axis in the Talairach space (in millimeters).

TE becomes longer and would be hard to correct because information concerning the local T_{2Cap}^* dynamic change (or blood partial oxygen saturation) is required. Although this effect might also affect the resting rCBF measurements, it is possible that the effect would average out if the rCBFs were assessed after sufficiently long resting periods.

Of interest, in comparison to gray matter, white matter does have the reverse sensitivity to TE because R_{2WM}^* is larger than $R_{2WM\_Cap}^*$. Although our estimation of these two values is not precise, it suggests that $R_{2WM\_Cap}^*$ is similar to that of arterial blood, i.e., white matter capillary blood is much more oxygenated than gray matter capillary blood. This is a surprising conclusion considering that the oxygen extraction fraction for white matter capillary blood is similar to that of gray matter (17,18) and that the partial oxygen pressure is similar for gray and white matter capillaries (19,20). Moreover, the rCBF corrected value was smaller than expected: 12 mL/100 g/min, whereas PET measurements ranged from 17 to 26 mL/100 g/min (17,21–23). Nonetheless, the rCBF white matter values should be corrected independently of those for gray matter.

Our suggested correction should not be viewed as an attempt to match the biophysical reality, but rather as a theory-driven empirical adjustment. More sophisticated two-compartment models should be used for the former, e.g., that of St Lawrence and Wang (6), although the two-compartment model may also be affected by the BOLD effect. Our contribution is a practical one and hence is simple. However, it should not be used for gray matter rCBF calculations when TE is $\geq$27 ms as linearity vanished beyond this point (Fig. 1b). Moreover, when applied to white matter rCBF values, the results were not credible because the calculated white matter blood flow was 12 mL/100 g/min instead of ~20 mL/100 g/min and the $R_{2WM\_Cap}^*$ of -4 s^{-1} is an unrealistic value. These unrealistic values may result from the δR_2^* correction, as our R_{2WM}^* measurement replicated previous measurements (24).

In conclusion, we propose a simple and practical correction to be used during rCBF calculations that accounts for TE effect in gray matter. It appears to improve the accuracy of rCBF values even for short TEs. Because the signal-to-noise ratio diminished and the sensitivity to the BOLD effect increased with increasing TE, it might be wise not to use our corrections for TEs > 24 ms. Finally, the correction should be done separately for white matter.

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